



Final Report

Unsteady RANS vs. LES

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John Gomes

Professor Xinyu Zhao

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Abstract

Computational power is the forefront of modern engineering. Now, simulations are almost a necessary precursor for physical testing. In many cases, this can save both time and money. Though, correct modeling approaches must be followed to get reliable results. In many industries, such as aerospace, automotive and renewable energy engineering, turbulent flows dominate. Computational Fluid Dynamics (CFD) has become a necessary tool for studying complex flows.

Two of the most widely used turbulent solving techniques are Unsteady Reynolds-Averaged Navier-Stokes (URANS) and Large-Eddy Simulation (LES). Both approaches attempt to solve the Navier-Stokes equations, but their process for solving these equations differs. URANS uses Reynolds decomposition of the Navier-Stokes equations, splitting an instantaneous flow into a time-averaged and fluctuating component. The time-averaged term is solved directly, while the fluctuating term is approximated with closure models, such as $k - \omega$ SST or $k - \epsilon$, resulting in a smoothed solution [1].

LES uses spatial filtering rather than time-averaging. This separates turbulent flow into distinct scales. LES directly resolves larger eddies, while approximating smaller sub-grid scale (SGS) eddies. Direct resolution of large turbulent flows yields a higher accuracy, particularly in complex flows, such as flow separation. It also means significantly higher computational cost when compared to URANS, especially when resolving the viscous sublayer in near-wall regions [1].

Ultimately, neither solving technique is inherently superior to the other; they represent a fundamental trade-off between physical accuracy and numerical efficiency. This report provides a comparative analysis of these two models, evaluating their mathematical differences and performance metrics such as accuracy, stability, and computational costs.

1. Introduction

Both URANS and LES are numerical techniques that aim to solve the Navier-Stokes equations. First, continuity can be shown as follows. Note that $\mathbf{u}$ represents the three-dimensional velocity vector field.

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

Next, the conservation of momentum is generalized as follows. Here, ρ and ν are fluid density and kinematic viscosity respectively.

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} \quad (2)$$

To understand the complexity of turbulence modeling we must break down how URANS and LES address the non-linearity of the Navier-Stokes equations. Since solving these equations directly is often too computationally challenging, both models rely on mathematical approximations. This paper will compare the theoretical framework for both methodologies and quantitatively compare URANS and LES by simulating tube flow over a cylindrical bluff body.

2. Governing Equations & Turbulence Modeling

2.1 URANS

Fundamentally, URANS is the separation of the flow into a time-averaged component, $\bar{u}_i$, and a fluctuating component, u'_i , which is modeled rather than solved directly. This is done using Reynolds Decomposition [2]. See the equation below for the Einstein representation of velocity decomposition.

$$u_i = \bar{u}_i + u'_i \quad (3)$$

This decomposition can also be used for the pressure term shown in equation 2.

$$p = \bar{p} + p' \quad (4)$$

Substituting the decomposed velocity and pressure back into the Navier-Stokes equations results in the Reynolds-Averaged Navier-Stokes equations [2]. Note that with this substitution, the continuity equation remains unchanged.

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{\partial (\overline{u'_i u'_j})}{\partial x_j} \quad (5)$$

The rightmost term is critical in URANS, this is known as the Reynolds Stress Tensor, also known as Reynolds stress. With the addition of this term, the system of equations is no longer closed. This is because the tensor contains undetermined, orthogonal fluctuations of momentum (u'_i and u'_j) [2].

2.1.1 URANS Closure Models

To solve the Reynolds-Averaged Navier-Stokes equations, a closure model must be used. The three most used closure models are $k - \epsilon$, $k - \omega$ and $k - \omega$ Shear Stress Transport (SST).

First, the $k - \epsilon$ closure model solves turbulent kinetic energy, k , and absolute turbulent dissipation rate, ϵ , by defining eddy viscosity as follows. Here, C_μ is an empirical constant-

$$\nu_t = \frac{C_\mu k^2}{\epsilon} \quad (6)$$

The transport equation for k describes the transfer of energy from the time-averaged velocity field to the fluctuating component. The equation below is for an incompressible fluid. Note that σ_k is the turbulent Prandtl number for k , and S is any source term [3].

$$\frac{\partial k}{\partial t} + \bar{u}_j \frac{\partial k}{\partial x_j} = P_k - \epsilon + \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \frac{\partial}{\partial x_j} \right] + S \quad (7)$$

Where turbulent production, P_k , is:

$$P_k = \nu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \frac{\partial \bar{u}_i}{\partial x_j} \quad (8)$$

In similar format, the turbulent dissipation rate transport equation is as follows. Here, σ_ϵ is the turbulent Prandtl number for dissipation rate.

$$\frac{\partial \epsilon}{\partial t} + \bar{u}_j \frac{\partial \epsilon}{\partial x_j} = C_{\epsilon 1} \frac{\epsilon}{k} P_k - C_{\epsilon 2} \frac{\epsilon^2}{k} + \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\nu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right] \quad (9)$$

See the table below for a summary of the empirical constants used in the standard $k - \epsilon$ closure model

Table 1: Empirical Constants for $k - \epsilon$ Closure Model [3]

C_μ	$C_{\epsilon 1}$	$C_{\epsilon 2}$	σ_k	σ_ϵ
0.09	1.44	1.92	1.0	1.3

Essentially, these equations work together to calculate local eddy viscosities by coupling turbulent energy and the eventual decay. This model works especially well for free-stream flow but struggles in near-wall conditions. Fundamentally, a no-slip boundary condition tells us velocity is zero, therefore turbulent kinetic energy must also be zero. Furthermore, at the wall, viscous shear stress is the highest, $\epsilon \neq 0$, thus a singularity within the destruction term [3].

$$\lim_{k \rightarrow 0} \left(C_{\epsilon 2} \frac{\epsilon^2}{k} \right) = \infty \quad (10)$$

This is where $k - \omega$ closure model becomes preferable. This closure model models turbulent kinetic energy with the specific turbulent dissipation rate, ω . Here, eddy viscosity is defined as:

$$\nu_t = \frac{k}{\omega} \quad (11)$$

Like the equation for absolute turbulent dissipation rate, specific turbulent dissipation rate is as follows. See Table 2 below for empirical constants, α , β and σ_ω [4].

$$\frac{\partial \omega}{\partial t} + \bar{u}_j \frac{\partial \omega}{\partial x_j} = \alpha \frac{\omega}{k} P_k - \beta \omega^2 + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_\omega \nu_t) \frac{\partial \omega}{\partial x_j} \right] \quad (12)$$

Table 2: Empirical Constants for $k - \omega$ Closure Model [4]

α	β	σ_ω
5/9	3/40	0.5

Here, the destruction term no longer has k in the denominator, thus no mathematical singularities near-wall. While there is still a k in the denominator of the production term, this is canceled when multiplied with P_k .

Though, this model is extremely sensitive in free-stream conditions. This is a limitation to the fact that the model was created specifically for near-wall flow. Specific turbulent dissipation does not naturally decay to a free-stream value, unlike absolute turbulent dissipation in $k - \epsilon$ [5]. With this, solution accuracy is strongly dependent on prescribed free stream dissipation.

F.R. Menter essentially combined the ideas of both closure models mentioned above to create the most popular URANS closure model, $k - \omega$ SST. The transport equation for dissipation is similar, but with a new cross-diffusion term [5].

$$\frac{\partial \omega}{\partial t} + \bar{u}_j \frac{\partial \omega}{\partial x_j} = \tilde{\alpha} \frac{\omega}{k} P_k - \tilde{\beta} \omega^2 + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_\omega \nu_t) \frac{\partial \omega}{\partial x_j} \right] + 2(1 - F_1) \sigma_{\omega 2} \frac{1}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \quad (13)$$

Here, F_1 equals one when in the near-wall region and equals zero far from the surface. [5]. This allows for the benefits of both mentioned closure models, allowing for resolution near-wall, without freestream dependency. This model also uses the following blending function in transitional regions. This reduces the chance of neighboring cells having substantially different results.

2.2 LES

LES separates flow spatially. With this, it directly solves large-eddy effects, $\tilde{u}_i$, and models the effects of small eddies, u_i'' . This is done using spatial filtering rather than time-averaging [1]. See the equation below for the Einstein representation of velocity decomposition.

$$u_i = \tilde{u}_i + u_i'' \quad (14)$$

This decomposition can be used for pressure as well.

$$p = \tilde{p} + p'' \quad (15)$$

Substituting the decomposed velocity and pressure back into the Navier-Stokes equations, results in the Filtered Navier-Stokes equations [1]. Note that with this substitution, the continuity equation remains unchanged.

$$\frac{\partial \tilde{u}_i}{\partial t} + \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \tilde{p}}{\partial x_i} + \nu \frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j} - \frac{\partial(\tau_{ij})}{\partial x_j} \quad (16)$$

The rightmost term is known as the Subgrid-Scale (SGS) stress tensor. Similarly to URANS, this is required to close the system of equations. Note that here, τ_{ij} can be written as the difference between the exact momentum transfer and the solved values.

$$\tau_{ij} = \widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j \quad (17)$$

The grid filter, Δ , is typically used to determine which eddies will be solved directly, and which will be modeled [7]. Here, V_c is the volume of a given cell.

$$\Delta = V_c^{1/3} \quad (18)$$

This formulation for grid filtering relies on the premise that the computational cells are uniform, meaning they are cubic, or near-cubic cells. In regions of high skewness or aspect ratio, the a simple volumetric filter fails. With this, any turbulent eddy larger than the spatial filter is explicitly computed by the resolved Navier-Stokes equations, while the dissipative effects of any eddy smaller must be approximated [7]. This is why the mesh density and uniformity for LES must be rigorously verified when compared to URANS, as it dictates the amount of modeling rather than computing.

2.2.1 LES SGS Models

Two popular SGS models in LES are the Smagorinsky-Lilly Model, and the Wall-Adapting Local Eddy-Viscosity (WALE) model. First, the Smagorinsky model assumes the eddy viscosity, ν_{sgs} , is dictated by the size of a given cell, along with the magnitude of the strain rate tensor, S . Here, C_s is an empirical constant, typically between 0.1 and 0.2.

$$\nu_{sgs} = (C_s \Delta)^2 |\tilde{S}| \quad (19)$$

Where:

$$\tilde{S} = \frac{1}{2} \left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} \right) \quad (20)$$

Similarly to the $k - \epsilon$ closure model, this model struggles due to no-slip wall conditions. This is because the turbulent viscosity relies on the resolved strain rate tensor. It inaccurately overpredicts turbulent viscosity at the wall. Here, turbulent fluctuations should approach zero. To suppress the erroneous turbulence at the wall, damping functions are often required, compromising the solution accuracy [7].

Rather than calculating eddy viscosity based on strain rate, WALE uses resolved strain rate and vorticity. The sub-grid viscosity is as follows. Here, C_w is a WALE constant, typically $C_w = 0.325$ [8].

$$\nu_{sgs} = (C_w \Delta)^2 \frac{(S_{ij}^d S_{ij}^d)^{3/2}}{(\tilde{S}_{ij} \tilde{S}_{ij})^{5/2} + (S_{ij}^d S_{ij}^d)^{5/4}} \quad (21)$$

Here, S_{ij}^d is the traceless symmetric part of the square of the velocity gradient tensor. By using this, the WALE model isolates regions of pure laminar shear from regions of complex turbulent stretching and rotation [8]. Essentially, this acts as a mathematical filter, which allows WALE to predict smoothness along the walls, and detect turbulence where appropriate, such as in the free stream. With the equation above, turbulent viscosity can scale to zero, giving a more physically accurate answer. This is particularly useful for capturing separation about a bluff body, as shown in the following example.

3. Exemplifying URANS vs. LES

3.1 Model Setup

To compare both modeling techniques, I created a simple example of flow through a tube, with a cylindrical bluff body downstream of the inlet. See the figure below for the design parameters of this problem.

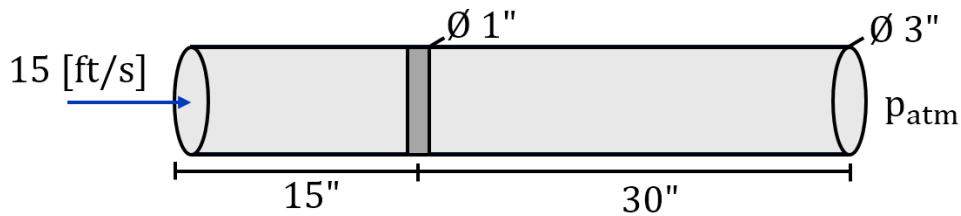


Figure 1: Example Problem Setup

This problem includes a 45-inch pipe, with a blockage 15 inches downstream from the inlet. There is an inlet velocity of 15 [ft/s], with an atmospheric pressure outlet. The general workflow for solving this problem is as follows.

Siemens NX → Siemens StarCCM+ → Ansys Fluent → CFDPost

3.2 Meshing

An Airsolid modeling approach was used to create a model for this problem. This is shown in the figure below.

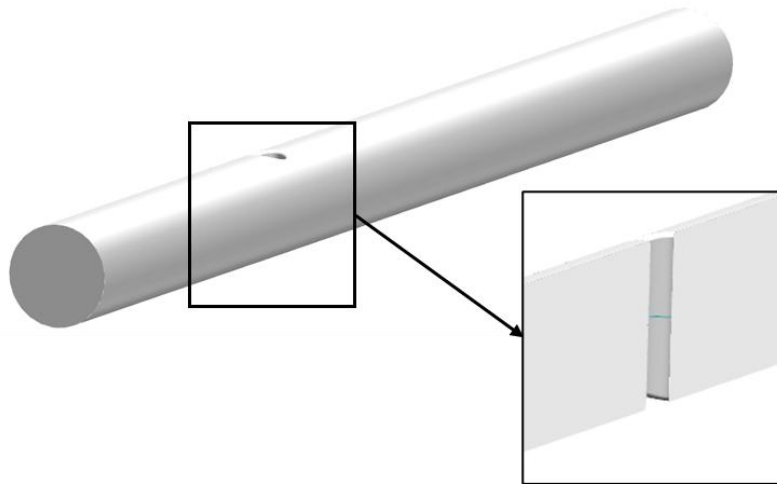


Figure 2: Pipe Airsolid

From this, a mesh was discretized in StarCCM+, using an automated polyhedral mesh. To satisfy the spatial resolution requirement of LES. As mentioned above, the accuracy of the model is dependent on the volume of each cell. This mesh also uses a prism layer mesh at the boundaries for increased accuracy near-wall. The figure below shows the inlet of the pipe. On the right, an example of the prism layer mesh is shown.

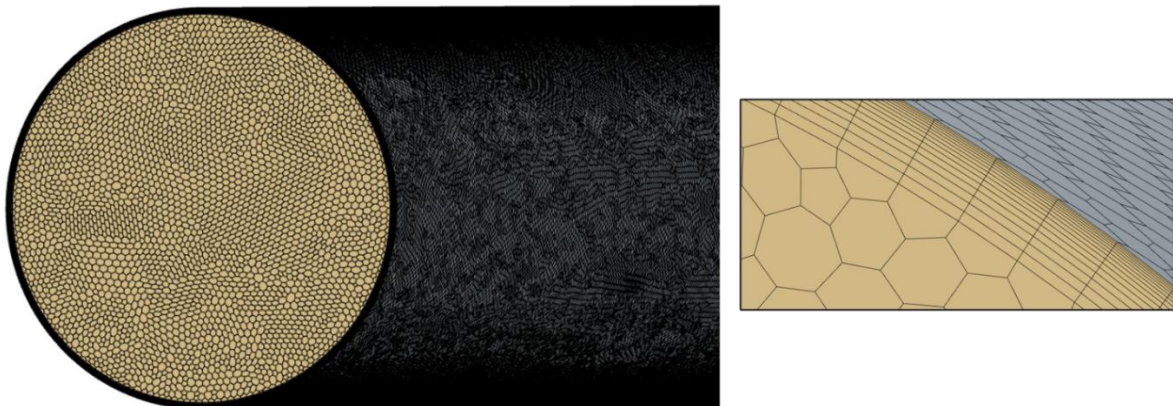


Figure 3: Pipe Mesh

This mesh is 6,147,897 cells, 31,785,436 faces in total.

3.3 Solving and Convergence

For both URANS and LES, second-order implicit formulations were used for the utmost accuracy. For URANS, the $k - \omega$ SST closure model was used, and for LES, WALE. As mentioned, these are some of the most widely accepted models by today's standards.

To validate this model, the following criteria were targeted: $y^+ \leq 1$, net mass flux error of 0.1% or less, and stable residuals at or under $1e-4$ for continuity and momentum equations, and $1e-7$ for the energy equation. These criteria, which are industry standards, were put in place to ensure a stable and physically accurate solution. See the equation below used for y^+ .

$$y^+ = \frac{u_T y}{\nu} \quad (22)$$

Expectedly, the y^+ value will be highest near the blockage, but even here, $y^+ \ll 1$.

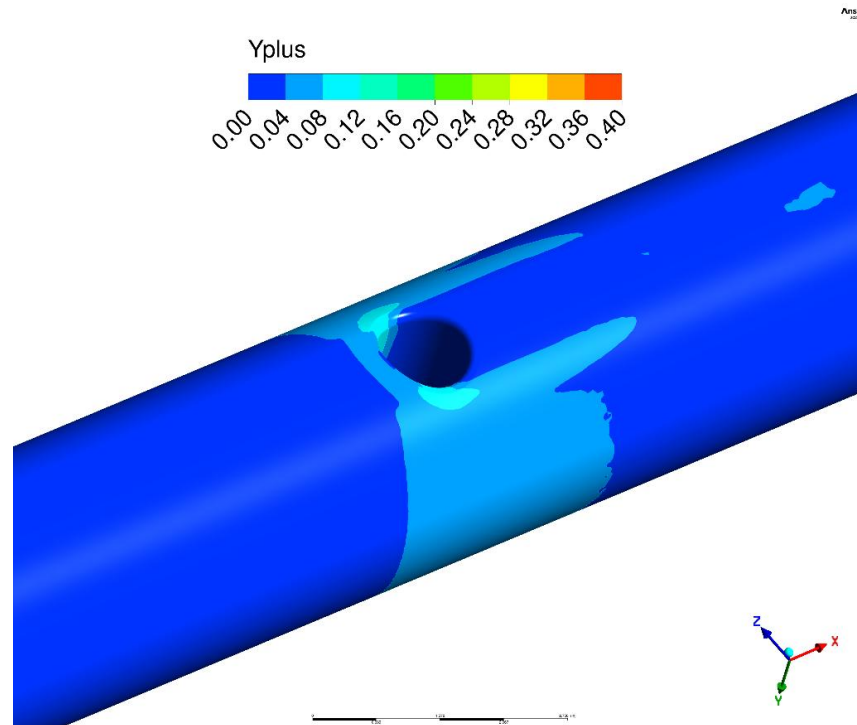


Figure 4: Pipe Wall y^+

While all the convergence criteria were met for both URANS and LES, it took roughly 1000 iterations to achieve this with URANS, while LES took about 10,000 iterations ($\Delta t = 0.0005\text{ s}$). URANS converged within hours, whereas LES required an overnight computation. Considering the simplicity of this model, it shows why many industries must choose URANS over LES for its computational efficiency.

3.4 Velocity Contours

The following plot shows a comparison of velocity around the blockage for both LES and URANS.

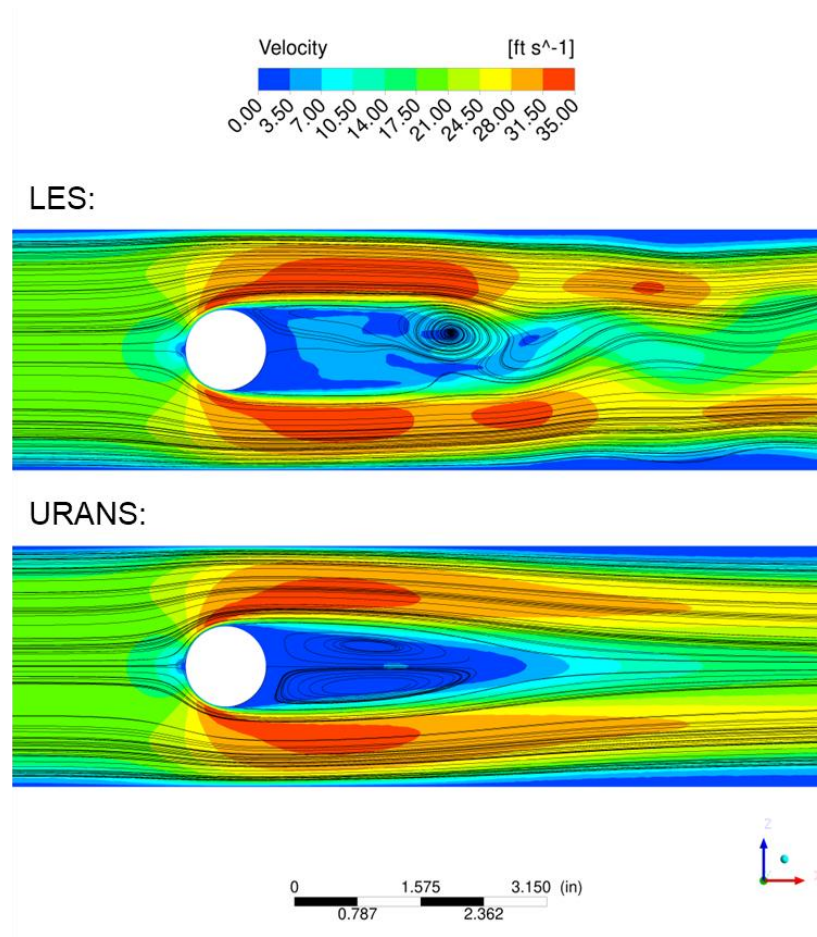


Figure 5: LES vs. URANS Velocity Comparison

The LES contour (top) captures a highly detailed transient behavior of the flow. Here, vortex shedding across the bluff body is evident. The black streamlines illustrate a strong recirculation zone aft of the blockage. This is because of the direct calculation of large energy containing eddies, which captures the oscillatory behavior of the flow as it moves downstream.

The URANS contour (bottom) shows a symmetrical flow field, a result of numerical smoothing. While $k - \omega$ SST is widely considered the best closure model for URANS, it still has its faults. It overpredicts high turbulent viscosity in the free stream, hence why there is no oscillatory behavior in the flow. Despite being a transient solution, it has fallen into a quasi-steady solution. There are still recirculation zones aft of the blockage, but they are severely misshapen.

Conclusion

This comparative study has proven that neither URANS nor LES are superior in every application. It is a fundamental tradeoff of computational cost and physical accuracy. For example, URANS is currently preferable for rapid design iteration. URANS modeling techniques, such as $k - \omega$ SST, provide a computationally efficient, robust time-averaged approximation of the flow field. If the goal is to study bulk flow properties, URANS and LES yield practically the same results.

On the other hand, LES provides the utmost physical accuracy. By directly resolving energy-dominating eddies, LES can capture more of the transient flow effects. Ultimately, it is up to the CFD practitioner to determine the balance of physical accuracy and computational cost to determine which model is appropriate. As computational resources continue to improve, the applicability of LES will increase, allowing for more industry adoption. The results shown in this simple example give much more insight into the flow and would be much preferable if it was not such a computational burden.

References

- [1] Pope, S. B. (2000). *Turbulent Flows*. Cambridge University Press.
- [2] Wilcox, D. C. (2006). *Turbulence Modeling for CFD*. DCW Industries.
- [3] ANSYS Fluent Inc. (n.d.). *Standard, RNG, and realizable k-epsilon models*. University of Washington. <https://courses.washington.edu/mengr544/handouts-10/Fluent-k-epsilon.pdf>
- [4] SimScale. (n.d.). *k-omega SST*. SimScale Documentation. Retrieved April 22, 2026, from <https://www.simscale.com/docs/simulation-setup/global-settings/k-omega-sst/>
- [5] Menter, F. R. (1994). Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA Journal*, 32(8), 1598–1605.
- [6] CFD-Online. (n.d.). *Turbulence modeling*. CFD-Online Wiki. Retrieved April 22, 2026, from https://www.cfd-online.com/Wiki/Turbulence_modeling
- [7] Blazek, J. (2015). *Turbulence Modeling*. In *Computational Fluid Dynamics: Principles and Applications*. Butterworth-Heinemann.
- [8] Nicoud, F., & Ducros, F. (1999). Subgrid-scale stress modelling based on the square of the velocity gradient tensor. *Flow, Turbulence and Combustion*, 62(3), 183–200.
- [9] Bechmann, A., Sørensen, N. N., & Johansen, J. (2011). *On the Accuracy of uRANS and LES-Based CFD Modeling Approaches for Rotor and Wake Aerodynamics of the (New) MEXICO Wind Turbine Rotor Phase-III*. International Energy Agency (IEA) Mexnext Project Report.
- [10] Wang, M., & Zierke, W. C. (2005). *Comparison of Surface and Near Wake Flow Quantities Using URANS and LES Simulations*. Naval Undersea Warfare Center Division, Newport, RI.
- [11] Giusti, A., Mastorakos, E., Magri, L., & Jones, W. P. (2016). *A Comparison of URANS and LES for Soot Predictions in an Aero-Engine Model Combustor*. Proceedings of the Combustion Institute.